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Course: Project Course B

Project Title: Student Performance Predictor

1. Introduction

I chose the Student Performance dataset because I wanted to see how machine learning can predict a student's final grade based on their study habits and absences. This project is very useful because it helps schools find struggling students early and give them help before the final exams.

2. Dataset Description

The dataset has information about students, like their total absences, study hours, and past grades. The target column is the final grade, which has 5 different classes. Before training, I cleaned the data, scaled the numbers, and split it into 80% for training the models and 20% for testing them.

3. Models Used

I trained three models: Decision Tree, Random Forest, and XGBoost. To compare them, I checked their accuracy, precision, recall, and F1-score. Decision Tree was the worst with 89.35% accuracy. Random Forest was the best model because it got the highest accuracy of 92.48% and an F1-score of 92.14%.

4. Neural Network

I built a Sequential neural network using Keras with three layers (128, 64, and 32 neurons). I also added Dropout layers so the model does not just memorize the training data. I trained it with a maximum of 150 epochs, and the early stopping callback stopped the training at 120 epochs. It reached 90.81% accuracy and a 90.36% F1-score on the test data.

5. Results & Comparison

When I compared all the trained models together, Random Forest performed the best. Here is the summary table of all the results:

Model	Accuracy	Precision	F1-Score	Recall
Random Forest	92.48%	92.64%	92.14%	92.48%
XGBoost	92.06%	92.21%	91.71%	92.06%
Neural Network	90.81%	90.95%	90.36%	90.81%
Decision Tree	89.35%	89.11%	89.21%	89.35%

XGBoost was very close to Random Forest, and the Neural Network came in third place. Even though neural networks are very smart, for this kind of table data, tree models like Random Forest work better and are much faster to train.

6. What I Learned

From this project, I learned that big and complex models like neural networks are not always the best choice for every dataset. Simple table data usually gets better results from tree algorithms. I also learned that cleaning and preparing the data correctly is the most important step to make any model work well.